Two permutations with equal characteristic value admit representatives meeting exactly in the certificates

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Editorial note. The counted statement is the author's labelled environment at source0.tex lines 242-244 and its proof environment opens at line 245; the pairing is by adjacency and the retained proof range is the whole environment.

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Editorial scope. One original result of the pinned author manuscript, counted once: representatives meeting exactly in the certificates for the permutation case (author label lem:2-int-coset-sym).

Allowed standard prerequisites (editorial).

- 2-intersecting family of permutations. A family of permutations such that any two of its members agree in at least two positions.. Its characteristic function is the object of the statement.
- Certificate of a permutation. The set of pairs on which a permutation is fixed, used to name the members of the family by a common core.. They are the sets whose intersection is prescribed in the conclusion.

- C(f). The smallest size of a certificate common to the family's members, bounded in the hypothesis.. It is the bound that makes the construction possible.

Lemma 0.1. *Let $f\colon S_n \to \{0,1\}$ be the characteristic function of a 2-intersecting family, and suppose that $C(f) \leq n-2$. If $f(\alpha) = f(\beta) = 1$ and C_α, C_β are minimum certificates for α, β (respectively), then $|C_\alpha \cap C_\beta| \geq 2$.*

Proof. We will show that there exist $\alpha' \in S_n$ satisfying C_α and $\beta' \in S_n$ satisfying C_β such that $\alpha' \cap \beta' = C_\alpha \cap C_\beta$, where we identify α', β' with their certificate representations. Since $f(\alpha') = f(\beta') = 1$ and f is the characteristic function of a 2-intersecting family, we have $|\alpha' \cap \beta'| \geq 2$, and so $|C_\alpha \cap C_\beta| \geq 2$.

We start by constructing α' satisfying C_α such that $\alpha' \cap C_\beta = C_\alpha \cap C_\beta$. Let $C_\alpha = \{(i_1, j_1), \ldots, (i_m, j_m)\}$, where $m \leq n-2$, let $i'_1, \ldots, i'_{n-m}$ be the i-indices not mentioned in C_α, and let $j'_1, \ldots, j'_{n-m}$ be the j-indices not mentioned in C_α.

For each i'_s, there is at most one j'_t such that $(i'_s, j'_t) \in C_\beta$. We can therefore arrange the indices in such a way that if $(i'_s, j'_t) \in C_\beta$ then $s = t$. We can now define α' explicitly:

$$\alpha'(i_1) = j_1, \ldots, \alpha'(i_m) = j_m, \alpha'(i'_1) = j'_2, \ldots, \alpha'(i'_{n-m-1}) = j'_{n-m}, \alpha'(i'_{n-m}) = j'_1.$$

The same argument (replacing C_α, C_β by C_β, α') shows that we can find β' satisfying C_β such that $\alpha' \cap \beta' = \alpha' \cap C_\beta = C_\alpha \cap C_\beta$, completing the proof. □